VIETNAM NATIONAL UNIVERSITY – HOCHIMINH CITY
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CLASSIFICATION OF BADMINTON STROKES USING MOTION SENSOR TECHNOLOGY AND MACHINE LEARNING TECHNIQUES

BY

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Under the guidance and approval of the committee, and approved by its members, this thesis has been accepted in partial fulfillment of the requirements for the degree.

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Chairperson

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Committee member

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HONESTY DECLARATION

My name is Nguyen Le Anh Phu. I would like to declare that, apart from the acknowledged references, this thesis either does not use language, ideas, or other original material from anyone; or has not been previously submitted to any other educational and research programs or institutions. I fully understand that any writings in this thesis contradicted to the above statement will automatically lead to the rejection from the EE program at the International University – Vietnam National University Ho Chi Minh City.

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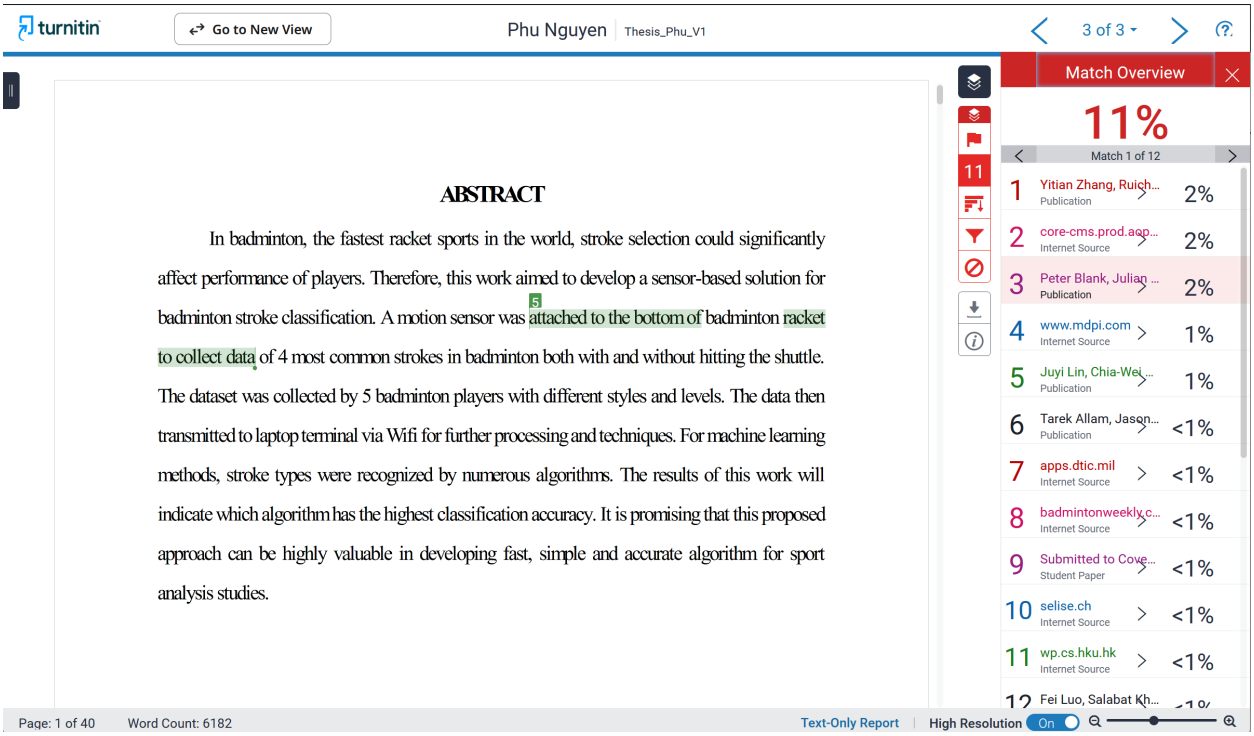
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TURNITIN DECLARATION

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ABBREVIATIONS AND NOTATIONS

IMU: Inertial Measurement Unit,

DPS: Degree Per Second

RMS: Root Mean Square,

TCP: Transmission Control Protocol,

CWA: Continuous Wave Accelerometer Data,

CSV: Comma Separated Value,

MCU: Microcontroller,

JSON: JavaScript Object Notation,

FFT: Fast Fourier Transform,

Std: Standard Deviation,

SVM: Support Vector Machine,

RF: Random Forest,

KNN: K-Nearest Neighbor

CNN: Convolutional Neural Network

ABSTRACT

In badminton, the fastest racket sports in the world, stroke selection could significantly affect performance of players. Therefore, this work aimed to develop a sensor-based solution for badminton stroke classification. A motion sensor was attached to the bottom of badminton racket to collect data of 4 most common strokes in badminton both with and without hitting the shuttle. The dataset was collected by 5 badminton players with different styles and levels. The data then transmitted to laptop terminal via Wi-Fi for further processing and techniques. For machine learning methods, stroke types were recognized by numerous algorithms. The results of this work will indicate which algorithm has the highest classification accuracy. It is promising that this proposed approach can be highly valuable in developing fast, simple, and accurate algorithm for sport analysis studies.

CHAPTER I

INTRODUCTION

In this chapter, 3 main sections were provided in total: Issue, Problem Statement, Goal and Objectives. In the Issue section, we will indicate the current issue. Next, the problems are stated in the Problem Statement part. Lastly, in the Goal and Objectives part, we present our expected goal with specific steps to reach it.

1.1 Issue

With roughly 200 million players worldwide according to the International Badminton Federation, Badminton is considered as one of the most famous sports [1]. Competitive Badminton matches are decided by tactics rather than players' ability. Therefore, Badminton stroke recognition can help improving training quality and analyzing tactics. Traditionally, analyzing players' performance to adjust their strategy is individually handled by coaches. This notational analysis method requires a lot of time, effort and attention from the coach which might not always be available. With the development of advanced technologies, stroke recognition research has been approached in two different methods which is by video-based or by inertial sensor-based. Since the video-based method has some significant defects: (1) high cost in developing the system, (2) difficulty in finding stable place to set up to maintain good illumination condition, (3) high computational complexity, (4) inflexible with dynamic backgrounds [2]. Hence, wearable sensor can overcome these drawbacks of the video-based method.

1.2 Motivation

The need for Badminton stroke classification is increasing. Therefore, sensor-based solutions are needed to enhance training and performance of Badminton players.

1.3 Goal and Objectives

Expected goal and the specific objectives to achieve that goal are presented in this part.

1.3.1 Goal

Badminton stroke classification system is developed to achieve fast and accurate classification. It is done by using inertial sensor technology and machine learning methods.

1.3.2 Objectives

- Objective 1: Data acquisition
  - Task 1: System platform
  - Task 2: Data collection
- Objective 2: Data preprocessing
  - Task 1: Noise reduction
  - Task 2: Segmentation
- Objective 3: Supervised learning
  - Task 1: Feature extraction and scaling
  - Task 2: Dimensionality reduction
  - Task 3: Classifier (SVM. KNN, RF)
- Objective 4: Deep learning (CNN-based architecture)

CHAPTER II

DESIGN SPECIFICATIONS AND STANDARDS

In this chapter, all the main components will be represented. Additionally, some of the most important features of these components will be mentioned.

2.1 Design Specifications and Standards

There were two main components in the hardware design. The first one was ESP 32 microcontroller. The second one was motion sensor GY-521 MPU 6050.

2.1.1 ESP 32 module



Figure 2.1: WEMOS LOLIN32 Lite Board

This MCU was picked due to its wireless communication functionality. In addition to Wi-Fi, the module also has Bluetooth as well as Bluetooth low energy function. It also has more IO pins and improved analog inputs. This ESP32 can be programmed under Arduino (C++) environment. The Lolin32 board includes a CH340G that converts USB to serial and allows

communication with the computer. Furthermore, the board has a 3.6V lithium-polymer battery interface which facilitates its wireless functionality. Lastly, another reason why I choose this board is because it has no header at the beginning, it suits my demand for building a wearable system design.

2.1.2 Motion sensor GY-521 MPU 6050



Figure 2.2: Gy-521 MPU 6050 module

The GY-521 MPU6050 includes a 3-axis accelerometer and a 3-axis gyroscope, which can be used to detect linear acceleration and rotational motion in six degrees of freedom. The sensor is designed to be compatible with popular microcontroller platforms like Arduino and PIC. It uses an I2C interface for communication with the microcontroller.

CHAPTER III

PROJECT MANAGEMENT

In this chapter, there are two main parts. The first part is about the budget and cost during the making of the project, the second part is about the planning and Gantt chart of the project.

3.1 Budget and Cost Management Plan

The cost of this project is reasonable due to the need of components backup. For those ESP32 module and the lithium-polymer batteries, their price covers the most expense. Table 3.1 shows component's prize.

Table 3.1: Cost management

Component Name	Total (unit)	Unit Price (VND)	Total Price (VND)
ESP32 NodeMCU LuaNode32 Module	1	138.000	138.000
ESP32 Lite V1.0.0 Module	1	120.000	120.000
LOLIN32 ESP32 Module	2	180.000	360.000
GY-521 MPU-6050	6	34.000	204.000
TP4056	6	12.000	72.000
Lithium Battery 300mAh – 1000maAh	6	70.000	420.000
Total			1.314.000

3.2 Project Schedule

The project started on 18/09/2023.

The deadline for report submission is on 17/01/2023.

The presentation for the project is on 27/02 – 02/03

Below is the project Gantt chart, which demonstrates the progress of this work.

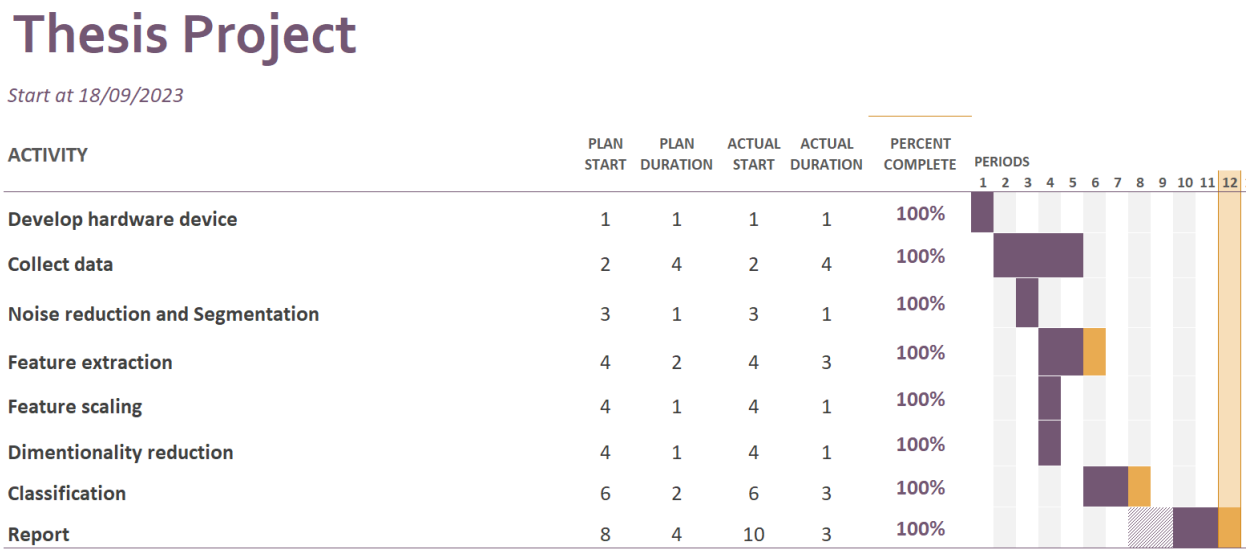


Figure 3.1: Gantt chart

CHAPTER IV

LITERATURE REVIEW

This chapter presents the background paper that can facilitate this research. There are two main parts in this chapter. The first part reviews the hardware design that could be used in this project. The second part focuses on the algorithm to classify the stroke. Specifically, supervised learning approaches and deep learning approaches are presented in the algorithm part. The purposes of these articles and knowledge that could be used in this project will be presented.

4.1 Hardware

Currently, the research in this field focuses mainly on video-based methods and sensor-based approaches. A video-based solution applied dense trajectories and trajectory aligned histogram of oriented gradients features. These features are computed inside bounding box around the players [3]. However, the limitation of this method is the camera resolution and dynamic blind spots which caused by the player's movement throughout the match [4].

Sensor-based solutions do not have the same problem as video-based approach. Recently, many researchers have used inertial sensors to detect and classify human physical activity [5, 6]. Xinye Zhang et al. [7] implemented ±16(g) accelerometer and ±2000(dps) gyroscope on wrists, angle, and waist of Badminton player to find the optimal number and inertial sensor placement. Yitian Yang et al. [8] proposed a sensor-based solution that using inertial sensor on the racket to classify 6 stroke motion in Badminton. A light sensor was combined with the racket to detect different grip mode of the players. Nurul Fathiah Ghazali et al. [2] implemented two Physilog 4 as their inertial sensor with ±16(g) accelerometer and ±2000(dps) gyroscope to classify 15 types of

strokes with 100Hz of sample rate. In another study by Mingyue Wu et al. [9], they proposed a classification solution in Tennis with extremely low sample rate at 50Hz to reduce computational load. Table 4.1 demonstrates the comparison of motion sensor choice, the placement of those sensors, detected activities, and the sample rate in data transmission.

Despite using various types of inertial sensors, most of the articles from Table 4.1 show that the range of their accelerometer and gyroscope remains. With the range of ±16(g) for the accelerometer, and ±2000(dps) for the gyroscope, it was proved to be appropriate for the task of sport activity recognition. Depending on the purpose of these articles, the number of inertial sensors is varied as well as the sensor placement of the player. In Badminton activity classification, two pivotal tactical factors can be classified including stroke and stance. In this research, only stroke classification is the focus. Both tasks require a minimal number of sensors for less impact on the players. For the stroke classification task, the bottom of the racket's grip was proved to be the optimal sensor position [4, 10, 8, 11].

Table 4.1: Existing Racket Sport Recognition Studies

References	Sport	Sensor	Sensor Placement	Detected Activities	Sample Rate (HZ)
[2]	Badminton	Acc: Physilog 4 (16g) Gy: Physilog 4 (2000dps)	Chest, Wrist	15	100
[4]	Badminton	Acc: Koala (16g) Gy: Koala (2000dps)	Racket	6	N/A
[7]	Badminton	Acc: 16g Gy: 2000dps	Wrists, Ankle, Waist	9	50
[8]	Badminton	N/A	Racket	6	50
[9]	Tennis	Acc: BMI160 (16g) Gy: BMI160 (2000dps)	Wrist	5	N/A
[10]	Badminton	Acc: MPU6050 (16g) Gy: MPU6050 (2000dps)	Racket	5	N/A
[11]	Badminton	Acc: Axivity AX6 (16g) Gy: Axivity AX6 (2000dps)	Racket, Wrist, Upper Arm	7	100
[12]	Badminton	Acc: Shimmer3 (16g) Gy: Shimmer3 (2000dps)	Wrist, Palm, Legs	12	512
[13]	Badminton	Acc: Shimmer3 (16g) Gy: Shimmer3 (2000dps)	Wrist, Palm, Legs	6	512
[14]	Badminton	Acc: Axivity AX6 (16g) Gy: Axivity AX6 (2000dps)	N/A	7	50

4.2 Algorithm

The algorithm part is divided into two parts: supervised learning and deep learning.

4.2.1 Supervised learning

There are two main machine learning classification approaches being applied by recent researchers. With supervised learning approaches, features must be extracted manually. Whereas in deep learning approaches, features can be extracted automatically. Therefore, it is very important to choose as much significant features as possible for supervised learning approaches.

Nurul Fathiah Ghazali et al. [2] choose Max, Min, Std, and Mean their features to implement in SVM and KNN. An overall of 83.4% accuracy was achieved without applying any additional technique before training the model. Yitian Zhang et al. [8] not just extract common statistical time-domain features: Mean, Variance, Covariance, Correlation Coefficient, Kurtosis, Skewness, Median, RMS, Mean Absolute Deviation, but also extract from two physical features: Movement Intensity and Signal Magnitude Area. With the largest size of extracted features among the listed papers, they apply feature scaling technique and then reduce feature dimension by using PCA. Finally, they achieved 95.36% overall accuracy with RF and 90.4% with KNN. Despite having different dataset, Yitian Zhang's approach has better classification performance compared to Nurul Fathiah Ghazali's approach. Some of the reasons are: (1) Yitian Zhang had more valuable extracted features, (2) Yitian Zhang applied feature scaling and dimension reduction technique to enhance their performance.

Mingyue Wu et al. [9] investigated multiple feature scaling techniques. Then they concluded that standard scaler produced the best result compared to other types of scalers. Standardization technique was used to standardize features to zero mean and unit variance [15]. This technique was applied and proved to be effective by numerous articles [9, 8].

Multiple techniques were proposed to choose the most appropriate out of all extracted features and to reduce the computational complexity [16]. Among them, principle component analysis (PCA) was mostly chosen and was proved to be effective [9, 8].

4.2.2 Deep learning

Multiple standard convolutional neural networks with diverse architectures and parameters were developed and proved to be exceptionally effective [12, 13, 11, 14, 17]. Tim Steels et al. [11] proposed that each stroke data has different optimal time frames ranging from 0.4s to 1.6s. Therefore, the paper came up with an ensemble learning technique as the solution for combining these different time frames to predict the final result. Additionally, to make the model more generalized with the limited dataset, Daniel Peralta et al. [14] proposed adding data augmentation after the segmentation step. The data augmentation step combined multiple methods: Jittering, Scaling, Time warping, and Permutation. Table 4.2 shows listed papers that applied supervised learning and deep learning.

Table 4.2: Comparison of Machine Learning Approaches for Stroke Classification

References	Feature extraction	Feature Scaling	Dimension Reduction	Classifier
Supervised Learning Approach				
[2]	Mean, Std, Min, Max	N/A	N/A	SVM (83.4%) KNN (78.5%)
[4]	Maximum, Minimum, Mean, Std, Root Mean Square, Crest Factor, Interquartile Range, Skewness, Kurtosis	N/A	N/A	Random Forrest 79.32%(general) 95.91%(personal)
[7]	Mean, Variance, Covariance, Skewness, Entropy, Energy	N/A	LDA	KNN (96%) SVM (95%)
[8]	Mean, Variance, Covariance, Correlation Coefficient, Kurtosis, Skewness, Median, RMS, Mean Absolute Deviation, Energy, Movement Intensity, Signal Magnitude Area	Standard Scaler	PCA	KNN (90.4%) RF (95.36%)
[9]	Time domain	Standard Scaler	PCA	SVM (90%)
[10]	Mean, Median, Std, Var, RMS, Highest Peak, Lowest Peak, Energy	N/A	N/A	KNN (58%) SVM (88.89%)
Deep Learning Approach				
[11]	CNN			
[12]	CNN			
[13]	Residual Network (CNN)			
[14]	CNN			

CHAPTER V

METHODOLOGY

The methodology is described in detail for each of the objectives. There are several steps to accomplish these objectives, which are illustrated in the form of a system diagram. The project's system diagram will be shown in Figure 5.1 below. Additionally, an evaluation section will be illustrated in order to explain all the metrics that are used to evaluate the models' performance.

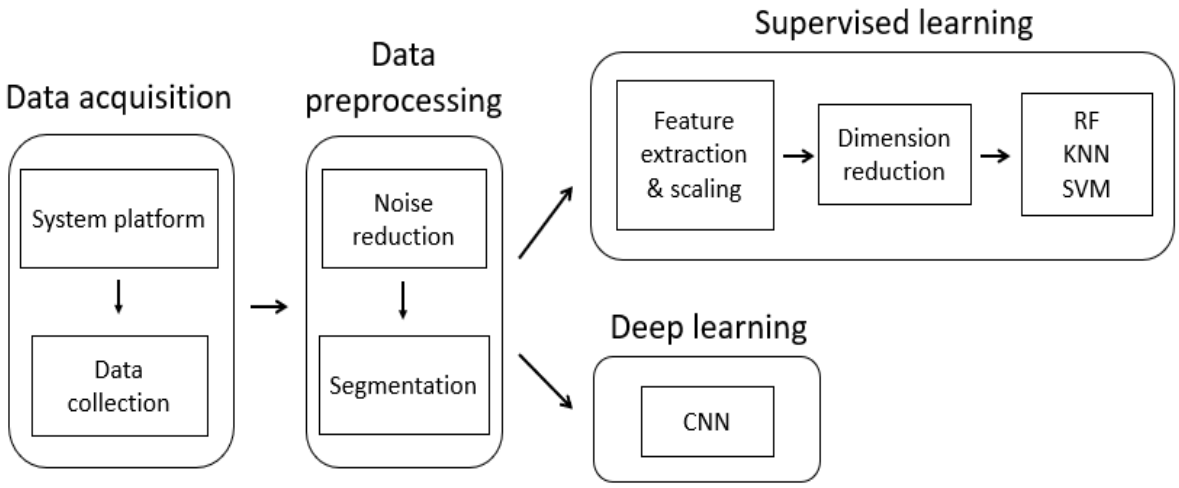


Figure 5.1: System diagram of this research

5.1 Data acquisition

In data acquisition step, there are two main steps: system platform, and data collection. System platform will demonstrate hardware design and inertial sensor selection. Data collection shows all the selected stroke for classification task. Additionally, methods that were used to perform these strokes will be discussed.

5.1.1 System platform

The complete device consisted of an ESP32 module as the system MCU (microcontroller), a motion sensor GY-521 MPU-6050, and a lithium-polymer battery that worked as the power supply for the system. Since the ESP32 and motion sensor MPU-6050 communicate through the I2C protocol, system platform is shown in Figure 5.2 below.

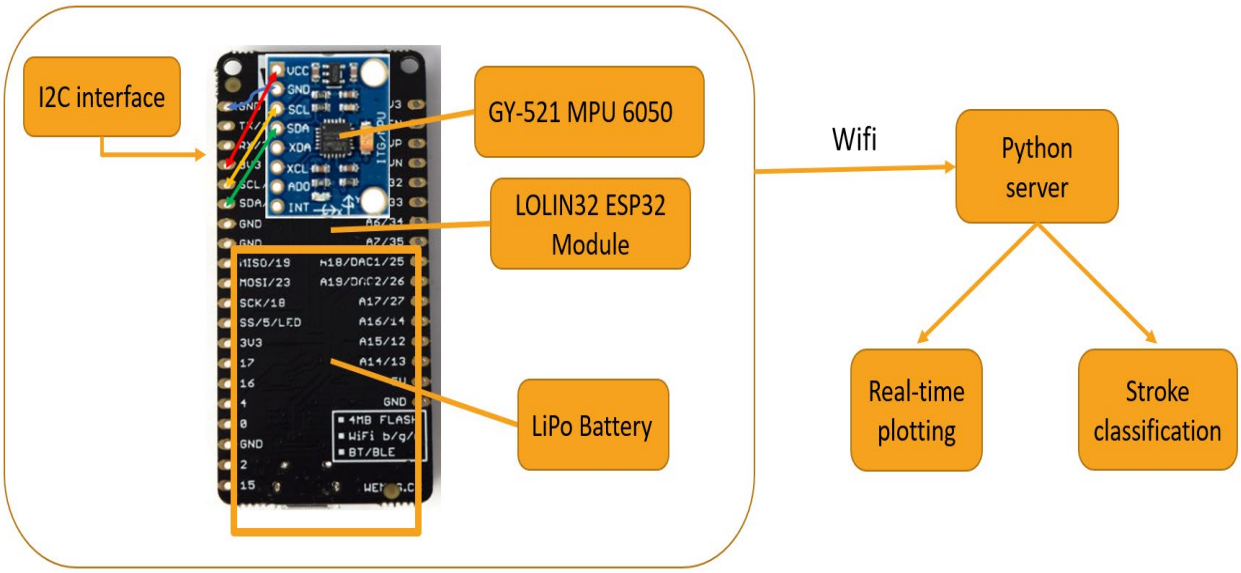


Figure 5.2: System platform of the project

From the simulated design, the practical one was built by soldering 4 pinouts. A double-sided tape was added below the MPU so that it could remain perpendicular to the surface of the ESP32 module. Lastly, a small lithium polymer battery with 500mAh capacity was assembled on the current board, next to the GY-521 MPU 6050. Next, the device was attached to the tail of the Badminton racket to perform its functionality.

The accelerometer in the GY-521 MPU6050 can measure up to four programmable full-scale ranges, which are ±2g, ±4g, ±8g, and ±16g. This means it can measure acceleration ranging

from ±2g up to ±16g, depending on the chosen full-scale range. The gyroscope in the GY-521 MPU6050 has a user-programmable full-scale range of ±250, ±500, ±1000, and ±2000(dps). This range determines the range at which the gyroscope can measure rotation rates or angular velocity. Since the action of the Badminton shot is extremely fast, all the scale settings must be set highest. This means, the accelerometer full-scale range was set to ±16g and the accelerometer full-scale range was set to ±2000(dps). Most of the articles investigated that this accelerometer and gyroscope range is appropriate and effective to recognize badminton stroke [2,4 7-14]. Data are transmitted from the device to the Python server via Wifi. The sampling frequency was set to 100Hz according to Tim Steel et al. [11] as the highest score on their weighted average accuracy.

In order to collect badminton stroke data, the JavaScript Object Notation (JSON) file format was chosen to store valuable training data. Briefly, JSON is a lightweight data-interchange format that is easy for humans to read and write, and easy for machines to parse and generate. Additionally, it is a language-independent format that is used to transmit data between numerous programming languages.

5.1.2 Data collection

After storing training data, the data collection step was set so as to collect suitable and essential data from the badminton racket. This research focuses on backcourt strokes. Hence, the four most basic strokes from the backcourt were chosen to perform the data collection including Clear, Slice, Drive, and Smash. Descriptions of the 4 strokes are listed as follows:

- **Clear**: played upward high in the air and landed at opponent's backcourt.
- **Slice**: played as close and fast as possible to the net.

- **Drive**: played on a horizontal or a bit downward path and down the court's sidelines.
- **Smash**: played downwards and straight with full energy and power.

All the strokes were performed in both fake type and real type. For fake shots, the player executed the stroke as simulating without the opponent and the shuttle. Meanwhile, for real shots, the shots were practically performed in a real game with the shuttle. Therefore, it was 8 categories in total to classify four real strokes and four fake strokes. The stroke visual description will be given in Figure 5.3

Afterward, data were collected by 9 badminton players, including three amateur and six advanced players. The players were asked to perform each stroke 100 times in a row and then switch to another stroke. The total of collected stroke data was roughly 9800 shots data from all 8 types of strokes. Table 5.1 shows the quantity of collected badminton strokes in the data collection stage. The dataset was then split into 3 sets including 60% training, 20% validating, and 20% testing.

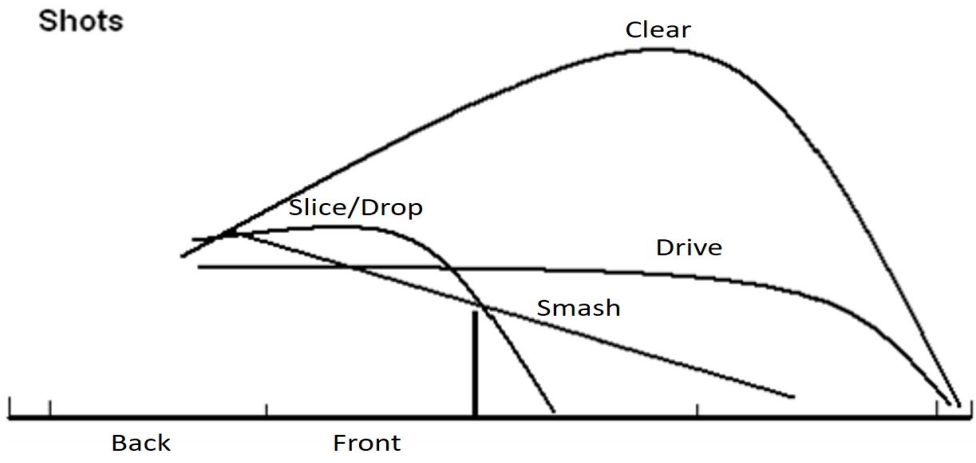


Figure 5.3: Badminton stroke visualization

Table 5.1: Quantity of each stroke type

Label	Stroke type	Quantity (shot)
Class 1	Clear fake	1167
Class 2	Slice fake	1162
Class 3	Drive fake	1189
Class 4	Smash fake	1171
Class 5	Clear real	1191
Class 6	Slice real	1217
Class 7	Drive real	1304
Class 8	Smash real	1283
Total		**9784**

5.2 Data preprocessing

In the preprocessing stage, raw data collected from the inertial sensor were filtered and segmented. Therefore, the data preprocessing step includes 2 tasks: noise reduction and segmentation.

5.2.1 Noise reduction

Diverse techniques of noise reduction were chosen by other articles to apply to their dataset. Zhelong Wang et al. [18] utilized wavelet transform function to filter out noise from acceleration and gyroscope signal. Mingyue Wu et al. [9] apply the Gaussian filter as their low

pass filter to smooth the signal. A Median filter was also proved to be effective in reducing the noise of motion sensor signals [12, 13, 19].

As motion sensor signal are collected in previous step. To reduce complex stochastic errors from collected inertial sensor data, a Gaussian filtering was applied [9]. The filter acts as a low-pass filter that smooths the signal but allows only lower frequencies to pass through it. The filtered signal is compared with raw signal in Figure 5.4

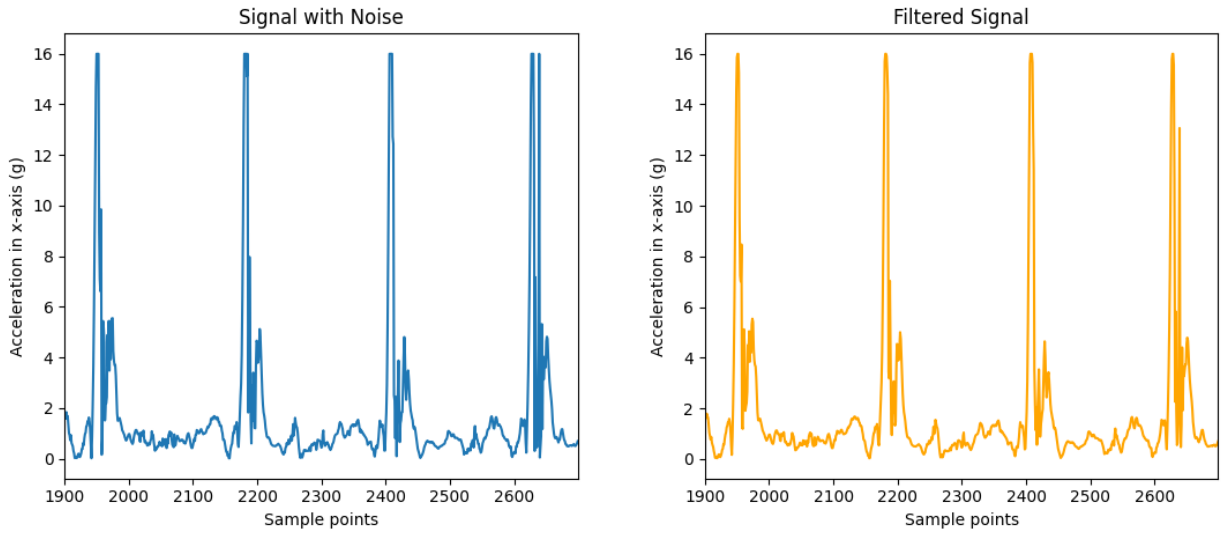


Figure 5.4: Raw signal (left) and Filtered signal (right)

5.2.2 Segmentation

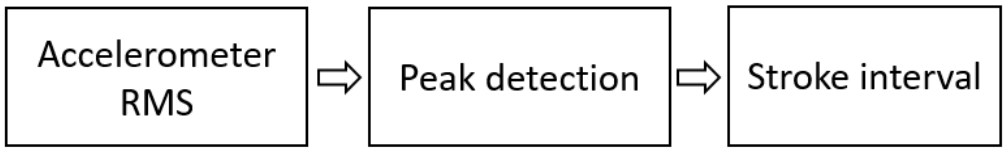


Figure 5.5: Window segmentation of badminton data

In order to segment data, Figure 5.5 illustrates three main steps: accelerometer RMS, peak detection, and stroke interval. Firstly, the RMS was computed from the accelerometer signal of X-axis $a_x(x_i)$, the Y-axis $a_y(x_i)$, Z-axis $a_z(x_i)$ with signal length L for each sample sequence, shown in equation (5.1).

$$Acc_{Rms}(x_i) = \sqrt{\frac{a_x(x_i)^2 + a_y(x_i)^2 + a_z(x_i)^2}{3}},\ 0 < i < L \qquad (5.1)$$

The signal is then used for peak detection by applying a threshold-based method [8, 20]. Lastly, when the peak frame was detected, 50 sample points before and after one stroke point were selected as the length of an observation window. By choosing the window size of 100 frame points, which corresponds to 1 second in the time interval. This makes sure that the stroke point can be located at the center of the window observation [18]. Figure 5.6 demonstrates that stroke points are at the center of the signal.

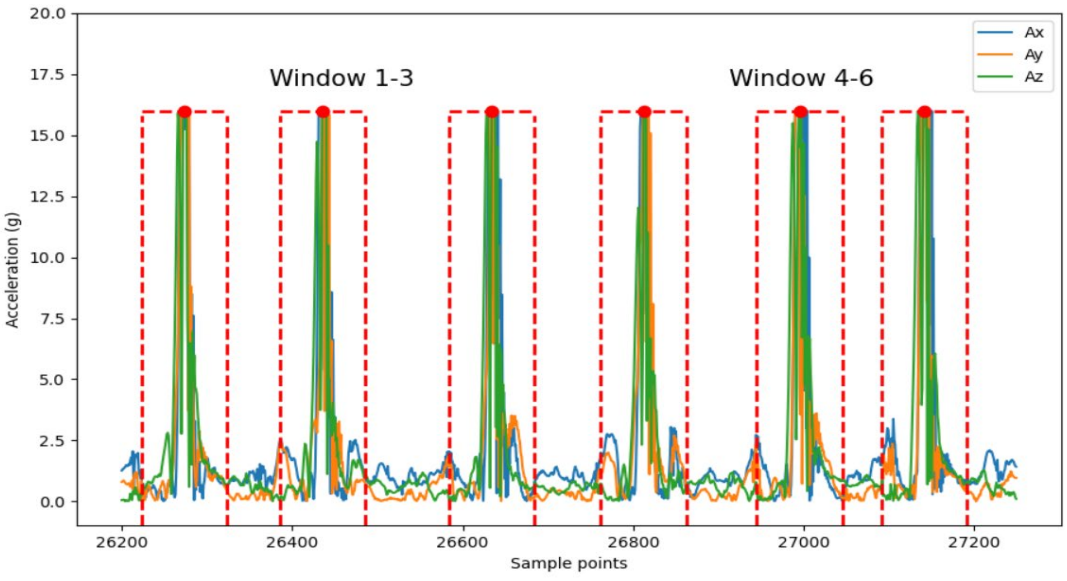


Figure 5.6: Window segmentation

5.3 Supervised learning

Supervised learning uses labeled datasets to train algorithms that predict outcomes accurately. The classification algorithm requires important information of signal as input in order to distinguish the differences between the classes. That information is called a feature. Therefore, the feature selection step is pivotal in supervised learning. Specifically, the classifier only yields decent performance with suitable features. Figure 5.7 shows the system diagram in supervised learning methods.

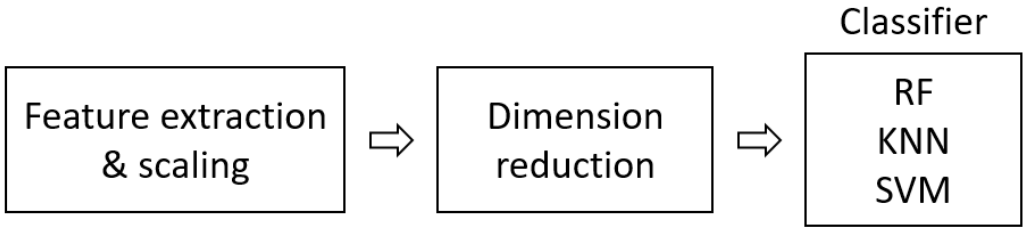


Figure 5.7: Feature selection in supervised learning

5.3.1 Feature extraction & scaling

Features are calculated from the stroke intervals which include the 6-axis inertial sensor and are summarized in Table 5. A total of 18 features are chosen from both time-domain and frequency-domain. Common statistical features include max, min, mean, root mean square, median, standard deviation, variance, skewness, kurtosis, covariance, correlation coefficient, crest factor, and interquartile range [4]. Besides, the chosen frequency-domain features are entropy and energy [7]. Finally, two physical features that combine movement intensity and signal magnitude area are chosen [8]. The final dimensionality of the feature vectors is 122 in total.

Since there are different ranges and units between different features, so the feature space needs to be standardized or normalized. In this research, the Z-score is used to standardize the features to zero mean and unit variance. The standardization equation is shown as follows.

$$f_{Standardized} = \frac{f-\mu}{\sigma} \qquad (5.2)$$

When f represents feature vector, μ is defined as the mean of f, and σ is defined as the variance of f.

Table 5.2: Feature definition

Feature	Description
Max	Maximum value of signal
Min	Minimum value of signal
Mean	Average value of a signal
Median	Middle value of a signal
Root Mean Square	Average "power" of a signal
Variance	Dispersion of a data set
Std	Spread of data around the mean
Skewness	Asymmetry of a distribution
Kurtosis	Tailedness of a distribution
Crest Factor	Farness of the peak stray from rms value
Interquartile Range	Middle half of the distribution
Covariance	Changes between the two variables, such that change in one variable is equal to change in another variable
Correlation Coefficient	Strength of a relationship between two variables (-1 to 1)
Mean Abs Deviation	Average difference between each value in the set and the mean
Movement Intensity	Euclidean norm of three axis signal
Signal Magnitude Area	Indirect estimation of energy consumption
Energy	Sum of power spectrum (frequency domain)
Entropy	Randomness or disorder of a signal in the frequency domain

5.3.2 Dimension reduction

When the dimensionality of features is too large, it will increase the computational complexity of the hardware system. Therefore, it is necessary to select the minimum number of important features to reduce the feature dimension. The chosen dimension reduction technique was PCA. This technique has been proven to be very effective [9, 8]. The feature dimension is 40 after implementing PCA technique.

5.3.3 Classifier

Different types of classifiers were compared because there is no best classifier for a specific classification task [21]. These classifiers include Random Forrest, K-Nearest Neighbor, and Support Vector Machine. In Support Vector Machine, linear kernel and radial based function (RBF) kernel were evaluated to find the optimal kernel. Each classifier has its own parameter to fit each task. The max_feature parameter in RF is selected as 6, which is rounded regarding the square root of the total selected features. n_estimator, max_depth parameters were performed grid search of 1 to 500 and 1 to 100 to find its optimal value in RF. For KNN, the k-nearest value was evaluated from 1 to 5. SVM was analyzed for regularization parameter C = $\in \{1,10,100,1000\}$. Furthermore, the gamma parameter in RBF SVM is also set to 'scale' mode as the optimal value for the classifier.

5.4 Deep learning

3 CNN architectures that are proposed in this part relating to 3 chosen papers, shown in Table 5.4. Indrajeet Ghosh et al. [12] proposed a standard CNN architecture. They implement 3 Conv1D layers as their convolution layers. The activation function for the three Con1D layers is

the ReLU or Rectified Linear Activation, which has been shown to perform well in neural networks. Then a fully connected layer was added after 'Flatten' layer to perform the classification task.

Sakorn Mekruksavanich et al. [13] proposed ResNet model. The model implements convolutional layers and residual connections to extract features proactively. Batch Normalization layer and ReLu layer was also added to this model. It was proved that these layers help reducing gradient vanishing and prevent overfitting issues.

Daniel Peralta et al. [14] proposed another standard CNN model. They implemented the Conv2D class from Keras as the first layer. As data passes through a deep neural network, the values can change, becoming either too large or too small. Batch normalization addresses this by normalizing the data, typically resulting in better outcomes. They also added 'Dropout' layers several times to prevent overfitting by ignoring random neurons during training. A "Flatten" layer is added between the Conv2D and Dense layers to serve as a connector. The Dense layer, a standard layer type in many neural networks, is used for our output layer. Their output layer consists of eight nodes, each corresponding to a potential outcome. The activation is "softmax". Softmax makes the output sum to 1 so that the output can be interpreted as probabilities. The model then makes its predictions based on which option has the greatest probability.

Table 5.3: CNN architecture comparison

Ref	CNN Architecture	Dropout rate	Batch size	Epoch
[12]	Conv1D (256, 15 × 1), ReLu Conv1D (128, 15 × 1), ReLu Conv1D (64, 9 × 1), ReLu Fully connected layer (8)	0.85	400	150
[13]	<u>Convolutional Block:</u> Conv1D, ReLu Batch Normalization MaxPooling <u>Residual Blocks × 3:</u> Conv1D, ReLu Batch Normalization Conv1D, Relu Batch Normalization <u>Outside Block:</u> Global Average Pooling Flatten Fully connected layer (8)	0.7	64	50
[14]	Conv2D (16, 2 × 2), ReLu Batch normalization Dropout Conv2D (32, 2 × 2), ReLu Dropout Flatten Fully connected layer (64) Fully connected layer (8)	0.3	64	100

5.5 Evaluation

The evaluation was done separately for supervised learning and deep learning. The purpose was to compare all predicted strokes with labeled strokes. The stroke type classification was based on the following metrics: Precision (5.3), Recall (5.4), F1-score (5.5), and Accuracy (5.6). These metrics evaluate all predicted strokes compared to all labeled strokes.

$$\text{Precision} = \frac{TP}{TP+FP} \tag{5.3}$$

$$\text{Recall} = \frac{TP}{TP+FN} \tag{5.4}$$

$$\text{F1-score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5.5}$$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{5.6}$$

When True Positive (TP) indicates when predicting a stroke as one class and it is indeed that class. False Positive (FP) indicates when predicting a stroke as one class but it is not that class. True Negative (TN) indicates when predicting a stroke as not that class and indeed it is not that class. False Negative (FP) indicates when predicting a stroke as not that class, but it is that class.

CHAPTER VI

RESULTS

All the results in data acquisition, supervised learning, and deep learning will be demonstrated in this chapter.

6.1 Data acquisition

In this first objective, the complete hardware design within the software developed in it will be illustrated.

6.1.1 System platform

The hardware device was attached to the Badminton racket in real life. A double-sided tape was added between the racket tail and the upper plane of the lithium-polymer battery to stick them together. Figure 6.1 illustrates the complete hardware design.

The design of the badminton racket handle was bigger to the end of the tail for keeping players' grip when playing a stroke. Based on that characteristic, a combination of three drawstrings was used so as to secure the connection of the device and the badminton racket.

6.1.2 Data collection

Since data were processed on a Python server, the visualization of how linear acceleration and angular velocity changed over time when the racket was shown in Figure 6.2 below. From the beginning to the frame of 2200, the racket was in rest position and all six motion sensor data maintained constant at nearly zero. After the frame of 2200, it was recognized that sensor data drastically fluctuated when the racket was in action for a stroke.



Figure 6.1 Final design of hardware system

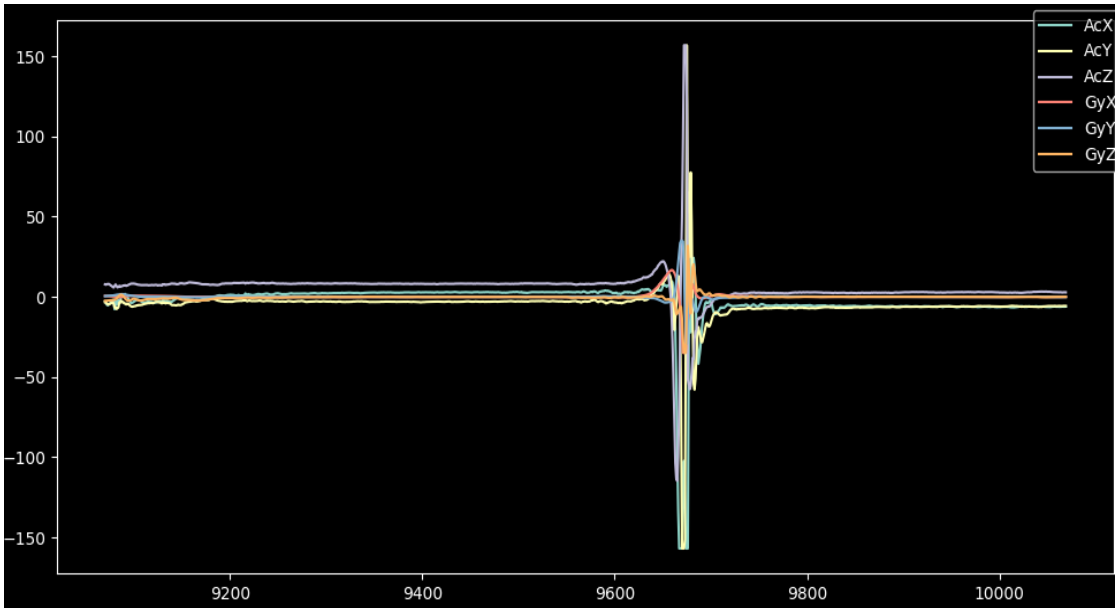


Figure 6.2 Racket motion plotting

6.2 Supervised learning

In the supervised learning part, the result of the paper evaluation and the investigation of different classifiers are demonstrated and discussed in Table 6.1. All 6 papers were investigated to find the best approach for this solution. Specifically, the paper's evaluation was examined by applying feature selection techniques and the classifier selection of those papers. Since each feature has its unique scale and unit, larger-scaled features can dominate the learning algorithm. Therefore, the feature scaling technique was applied to guarantee all features are on a comparable scale so that the learning algorithm treats all the features fairly. For feature selection, the more valuable features are being extracted the better the classifiers perform. In order to choose the best features out of a list of extracted features, PCA and LDA were implemented among the applied papers as dimensionality reduction techniques. By reducing feature dimensionality, redundant features are eliminated. Also, it helps reduces computational complexity and storage time so that it could prevent overfitting issues. The results from Table 7 show methods that apply feature scaling [9, 8] perform much better than those do not apply [2, 5, 6, 9]. Therefore, with enormous valuable features extracted, paper [7] performs the best with 97.2% accuracy. However, feature scaling does not necessarily improve the performance of RF since it is a tree-based algorithm. Other distance-based algorithms such as KNN or SVM performed much better when the feature scaling technique is applied.

Therefore, after evaluating all the listed papers, this research combines all features to extract from the collected dataset. Feature standardization techniques and PCA are also applied to the extracted features. After PCA, the feature dimension is 40. Finally, RF, KNN, and SVM with

linear (LIN) kernel and radial based function (RBF) kernel are implemented to perform the classification.

The max_features in the RF is selected as 6, which is rounded according to the square root of the feature dimension. The n_estimator and max_depth are selected through parameter traversal to 500 and 100. The accuracy corresponding to the n_estimator and max_depth is illustrated in Figure 6.3 and Figure 6.4. It can be found that with the increase of the n_estimator and max_depth, the accuracy is also increasing to a certain point before it become steady.

Table 6.1: Evaluation of different supervised learning papers

Ref	Num of Feature	Feature scaling	Dimension reduction	Paper results	Our Evaluation
[2]	48	N/A	N/A	SVM (83.40%) KNN (78.50%)	KNN (73.76%) SVM (72.07%)
[4]	54	N/A	N/A	RF 79.32%(general) 95.91%(personal)	RF (88.40%)
[7]	36	N/A	LDA	KNN (96%) SVM (95%)	SVM (65.78%) KNN (64.56%)
[8]	86	Standard Scaler	PCA	KNN (90.40%) RF (95.36%)	KNN (95.67%) RF (95.24%)
[9]	**120**	**Standard Scaler**	**PCA**	**SVM (90%)**	**SVM (97.20%)**
[10]	48	N/A	N/A	KNN (58%) SVM (88.89%)	SVM (68.29%) KNN (69.26%)

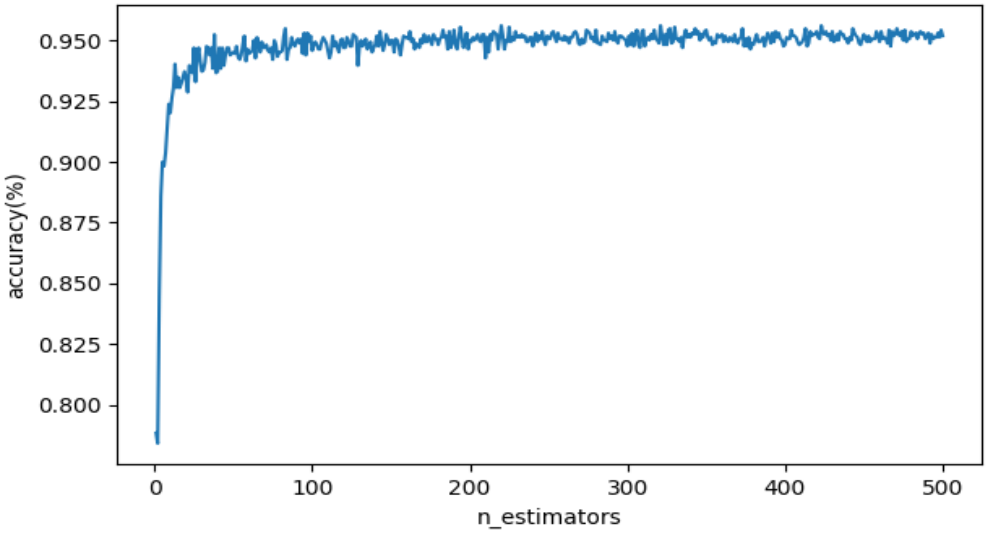


Figure 6.3: The effect of tree number on accuracy in RF model

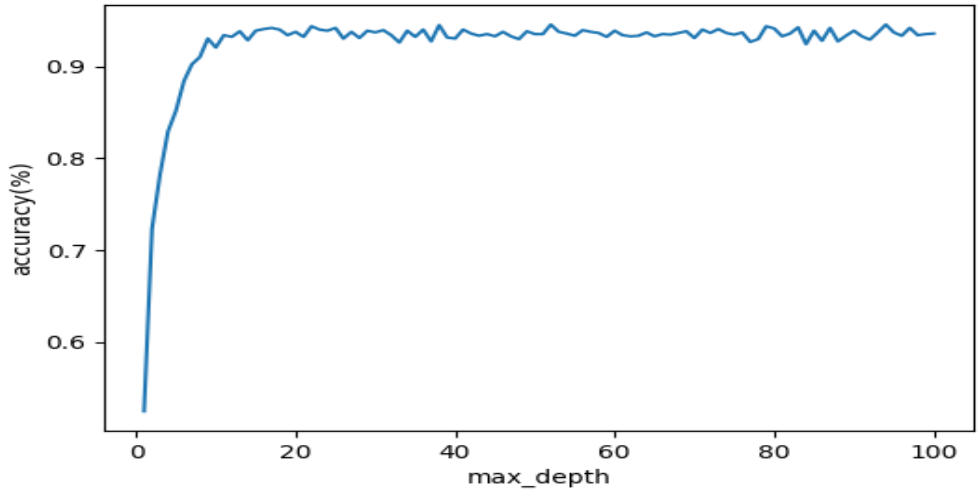


Figure 6.4: The effect of tree depth on accuracy in RF model

The following analysis of the influence of n_estimator and max_depth, shows that the accuracy will not improve when the max_depth reach a certain value. It is the same for the n_estimator.

Table 6.2: KNN accuracy with different value of k

Value of k	1	2	**3**	4	5
Accuracy (%)	95.60	94.93	**95.85**	95.36	95.79

The k value of KNN was analyzed from 1 to 5. It can be found that as k is increasing, the corresponding accuracy is not clearly improved. With k as 3, the model yields the highest accuracy at 95.85% and it is slightly better than the other k values.

Table 6.3: SVM accuracy with different value of C

Value of C		1	**10**	100	1000
Accuracy (%)	RBF	96.64	**97.43**	97.07	97.13
	LIN	94.26	93.59	93.65	93.59

Regularization parameter C was analyzed in both the linear kernel and radial based function kernel of the SVM classifier. For linear kernel, as C is increasing, the accuracy is not improving. Otherwise, as C is increasing in radial based function kernel, the accuracy is also increasing to a certain extent. It can be found that with C as 10, SVM with radial based function kernel performs its best at 97.43% accuracy.

Table 6.4: Classifiers comparison with 8 classes (F1-score)

Classifier		Clear fake	Slice fake	Drive fake	Smash fake	Clear real	Slice real	Drive real	Smash real	Average
SVM	**RBF**	**1.00**	**1.00**	**0.99**	**1.00**	**0.90**	**0.98**	**0.96**	**0.91**	**0.97**
	LIN	0.98	0.99	0.97	0.98	0.83	0.97	0.93	0.85	0.94
RF		0.99	1.00	0.99	0.99	0.82	0.98	0.93	0.84	0.94
KNN		0.99	1.00	0.98	0.99	0.88	0.99	0.96	0.88	0.96

Table 6.4 shows the F1 score of all classifiers in every class. It can be observed that RBF SVM performs the best with an overall classification rate of 97%. Clear real and Smash real are the two hardest classes to classify since it is practically similar in technique when executing. The RBF SVM performs better than all the others in these two classes with F1-score at Clear real (90%) and Smash real (91%). Four fake stroke classes are the easiest to classify with nearly 100% F1-score on each. Table 6.5 shows the confusion matrix of the RBF SVM. Labeled strokes and predicted strokes are compared with the corresponding rows and columns.

Table 6.5: Confusion matrix of RBF SVM

Class	C1	C2	C3	C4	C5	C6	C7	C8
C1	199	0	0	0	0	0	0	0
C2	0	214	0	0	0	0	0	0
C3	0	0	173	0	0	0	1	0
C4	0	0	0	186	0	0	0	0
C5	1	0	0	0	193	0	2	13
C6	0	0	0	0	1	225	2	0
C7	0	0	1	0	3	1	204	0
C8	0	0	0	0	15	1	1	204

6.3 Deep learning

In the deep learning part, the result of the paper evaluation and the investigation of different CNN architectures are demonstrated and discussed in Table 6.6. All 3 papers were investigated to find the best approach for classification. Specifically, the paper evaluation was examined by applying different types of convolution layers techniques and the number of convolution layers. Table 6.6 shows that the architecture of paper [14] performs the best with 97.87% accuracy, slightly better than paper [12] and paper [13] which are 97.01% and 97.13% respectively.

Training time per step is also a factor to be considered. With just 2.21 second, paper [14] had the shortest period in training each epoch. Therefore, it was the best model among three listed model in deep learning.

Table 6.6: Paper evaluation on 3 CNN models

Ref	Training time/step (s)	Paper accuracy	Our accuracy
[12]	5.32	86.27%	97.01%
[13]	6.63	98.00%	97.13%
[14]	**2.21**	**95.91%**	**97.87%**

To avoid overfitting problem, early stopping technique was applied to find the best number of epochs in training model. Figure 6.5 shows training loss compared to validation loss as epoch keep increase to 150 in training model of paper [12]. The CNN architecture of paper [14] performs better than the others at four real strokes, shown in Table 6.7. Four fake stroke classes are the easiest to classify with nearly 100% F1-score on each. Table 6.8 shows the confusion matrix of the paper [14] approach. Labeled strokes and predicted strokes are compared with the corresponding rows and columns.

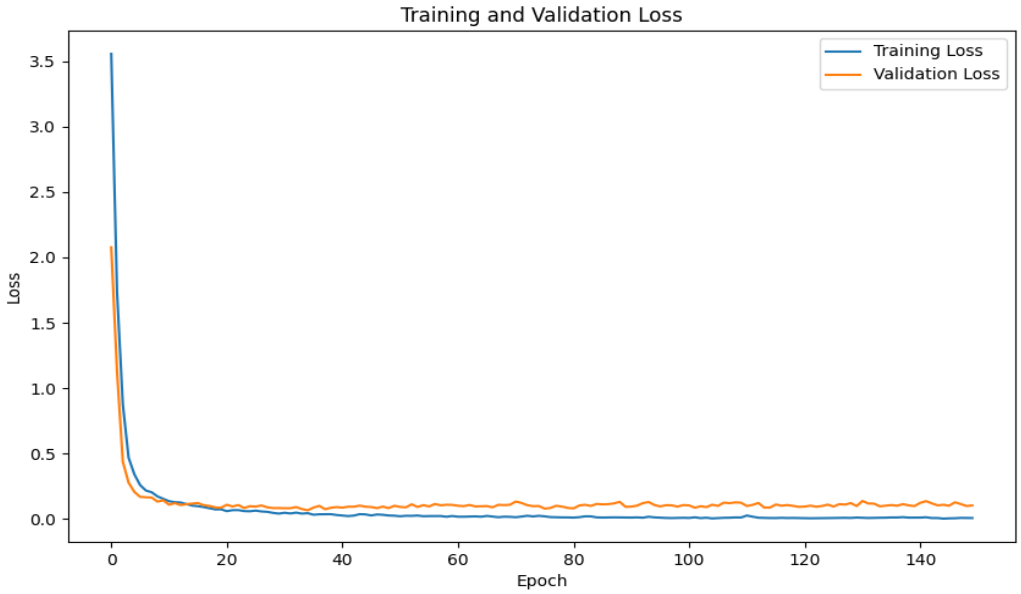


Figure 6.5: Training loss and validation loss in training paper [12] model

Table 6.7: CNN architectures comparison on 8 classes (F1-score)

Ref	Clear fake	Slice fake	Drive fake	Smash fake	Clear real	Slice real	Drive real	Smash real	Average
[12]	0.99	1.00	0.99	0.99	0.92	0.98	0.96	0.93	0.97
[13]	1.00	1.00	0.99	1.00	0.92	0.98	0.96	0.92	0.97
[14]	**1.00**	**1.00**	**0.99**	**1.00**	**0.94**	**0.99**	**0.98**	**0.94**	**0.98**

Table 6.8: Confusion matrix of the best CNN architecture [14]

Class	C1	C2	C3	C4	C5	C6	C7	C8
C1	199	0	0	0	0	0	0	0
C2	0	214	0	0	0	0	0	0
C3	0	0	174	0	0	0	0	0
C4	1	0	0	185	0	0	0	0
C5	1	0	0	0	197	0	2	10
C6	0	0	1	0	1	224	2	0
C7	0	0	1	0	2	0	205	1
C8	0	0	0	0	12	1	1	207

CHAPTER VII

CONCLUSION AND RECOMMENDATION

The conclusion and recommendation will be declared in this chapter.

7.1 Conclusion

In this research, motion sensor technology and machine learning technique were implemented to successfully classify 8 classes of Badminton stroke. A hardware design was developed to attach to the badminton racket and transmit data wirelessly to PC with the sampling rate at 100Hz. Throughout the project, a dataset of nearly 9800 stroke data was collected by a group of 5 Badminton players with different levels and experience. It was proved that either applying supervised learning method or deep learning technique, the system can achieve high accuracy at over 97%.

7.2 Recommendation

The purpose of this project was to provide a sensor-based solution to improve the self-analysis and training in badminton for players. Both supervised learning and deep learning were implemented to perform the task, the classification results were decent on the collected dataset. For more generalized dataset and new predicted data from new player, more strokes must be added to the selection as well as the quantity of players perform the stroke. Clear real and Smash real are the 2 most difficult stroke to classify. Since the techniques are almost the same, the only difference is the racket surface when the player hits the shuttle. Therefore, it required deeper analysis on the data to improve the classification result of these two classes.

CHAPTER VIII

BUSINESS, SOCIAL AND ETHICAL CONSIDERATIONS

In this chapter, the discussion resolves around the issue that the project is closely related to some fields as business, social and ethical.

8.1 Business

In business field, many badminton-related products are developed and produced to perform analyzing player's performance. In this research, hardware device was designed and machine learning approach was applied to perform the classification task. Our research product still needs to integrate and develop to be more efficient to users.

8.2 Social

In the social circumstance, this sensor-based solution can be implemented in the analysis of many sports. Also, with wireless technology and machine learning approach, many other fields such as fall detection can be implemented.

8.3 Ethical

On ethical perspective, the device is powered by a Lithium battery. Lithium battery is one of the most environmental-friendly energy options.

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APPENDICES